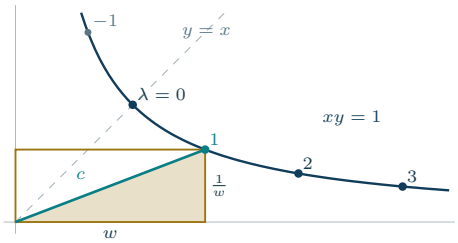
ALGEBRA ON ONE PAGE

Golden and metallic sides

A right triangle with legs w and $1/w$ has area $\frac{1}{2}$, and its sides carry the golden ratio and its metallic cousins.

1 / THE RULE AND THE PICTURE

The legs $(w,\ 1/w)$ are the coordinates of a point on the hyperbola $xy = 1$



A point (x, y) of the curve $xy = 1$ is a pair of reciprocals, so its coordinates are the legs w and $\frac{1}{w}$ of a right triangle of area $\frac{1}{2}$. The difference $x - y$ is the number λ, which gives $w^2 - \lambda w - 1 = 0$ and $w = \frac{\lambda + \sqrt{\lambda^2+4}}{2}$. The distance c from the origin follows from $x^2 + y^2 = (x - y)^2 + 2xy$, and $xy = 1$ turns this into

$$\boxed{c^2 = w^2 + w^{-2} = \lambda^2 + 2}$$

Sliding the point along the curve changes λ, and swapping the legs reflects it in $y = x$ and sends λ to $-\lambda$, the second root $-\frac{1}{w}$ of the quadratic. The marked points have $c = \sqrt{2}, \sqrt{3}, \sqrt{6}, \sqrt{11}$.

2 / THE METALLIC TRIANGLES

Each integer λ gives a family member, and the number under the root is $\lambda^2 + 4$

λ	name	w	$1/w$	$w + 1/w$	$c^2 = \lambda^2 + 2$
0		1	1	2	2
1	golden	$\frac{1+\sqrt{5}}{2} \approx 1.618$	$\frac{\sqrt{5}-1}{2} \approx 0.618$	$\sqrt{5}$	3
2	silver	$1 + \sqrt{2} \approx 2.414$	$\sqrt{2} - 1 \approx 0.414$	$2\sqrt{2}$	6
3	bronze	$\frac{3+\sqrt{13}}{2} \approx 3.303$	$\frac{\sqrt{13}-3}{2} \approx 0.303$	$\sqrt{13}$	11

The sum of the legs is $w + \frac{1}{w} = \sqrt{\lambda^2 + 4}$ and their difference is λ, so both are easy to read off. For bronze, $w^2 = \frac{11+3\sqrt{13}}{2}$ and $w^{-2} = \frac{11-3\sqrt{13}}{2}$ add to 11.

3 / WHY FIBONACCI APPEARS

The rule $w^2 = \lambda w + 1$ builds the powers of w one step at a time

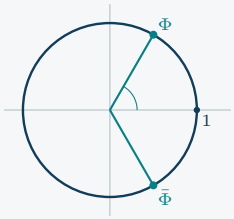
rule	case	powers of w and the numbers in them
$\varphi^2 = \varphi + 1$	$\lambda = 1$	$\varphi^3 = 2\varphi + 1,\ \varphi^4 = 3\varphi + 2,\ \varphi^5 = 5\varphi + 3$: Fibonacci $1, 1, 2, 3, 5, 8, \ldots$
$w^2 = 2w + 1$	$\lambda = 2$	$w^3 = 5w + 2,\ w^4 = 12w + 5$: Pell numbers $1, 2, 5, 12, 29, 70, \ldots$

Each power is the previous power times w, so the numbers in front obey $a_{n+1} = \lambda a_n + a_{n-1}$.
Ratios of neighbours then close in on w: $\frac{21}{13} = 1.615$ and $\frac{34}{21} = 1.619$ approach $\varphi = 1.618\ldots$, and $\frac{29}{12} = 2.417$ approaches $1 + \sqrt{2} = 2.414\ldots$.

4 / WHERE THE SQUARE ROOT COMES FROM

The same rule with imaginary λ puts w on the unit circle

The family $M_p = \frac{1+\sqrt{p}}{2}$ solves $M^2 - M = \frac{p-1}{4}$, so $p = 5$ gives $M^2 - M - 1 = 0$ and the root φ, and the roots are real while $p \geq 0$.
On the real curve write $w = e^t$, so that $\lambda = 2\sinh t$ and $c^2 = 2\cosh 2t$, with $\varphi = e^{0.4812}$.
For $p = -3$ the equation $M^2 - M + 1 = 0$ has the root $\Phi = \frac{1+\sqrt{-3}}{2}$, and $\Phi - \Phi^{-1} = \sqrt{-3}$ is an imaginary λ.
Writing $w = e^{i\theta}$ gives $\lambda = 2i\sin\theta$ and $c^2 = 2\cos 2\theta$, so $\theta = 60°$ gives $\lambda = i\sqrt{3}$ and $c^2 = -1$.
The hyperbola and the circle are two real views of one reciprocal rule, and the next sheet follows the circle.



ASIDE / TRY THESE

Check $(1 + \sqrt{2})(\sqrt{2} - 1) = 1$. For hypotenuse 2 take $\lambda = \sqrt{2}$: $w = \frac{\sqrt{2}+\sqrt{6}}{2} = 2\cos 15°$.

LINKS TO OTHER SHEETS

Next: the complex companion of the golden ratio. The silver case is the Pell sheet.

Convention. $\varphi = \frac{1+\sqrt{5}}{2}$; $w > 0$ solves $w - 1/w = \lambda$; c is the hypotenuse of the right triangle with legs w and $1/w$.